

Integration & Applications

ANSWER KEY



Basic antiderivatives (the power rule)

- 1 Find $\int x^4 dx$.
 $\frac{x^5}{5} + C$.
 - 2 Find $\int 3x^2 dx$.
 $x^3 + C$.
 - 3 Find $\int \sqrt{x} dx$.
 $\int x^{1/2} dx = \frac{2}{3}x^{3/2} + C$.
 - 4 Find $\int \frac{1}{\sqrt{x}} dx$.
 $\int x^{-1/2} dx = -\frac{1}{-1/2}x^{-1/2} + C = -\frac{1}{-1/2}x^{-1/2} + C = \frac{1}{1/2}x^{-1/2} + C = 2x^{1/2} + C = 2\sqrt{x} + C$.
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Antiderivatives of polynomials

- 5 Find $\int (4x^3 - 6x + 5) dx$.
 $x^4 - 3x^2 + 5x + C$.
 - 6 Find $\int (2x + 7) dx$.
 $x^2 + 7x + C$.
 - 7 Find $\int (x^2 - 4x + 3) dx$.
 $\frac{x^3}{3} - 2x^2 + 3x + C$.
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Antiderivatives of trigonometric and exponential functions

- 8 Find $\int \cos x dx$.
 $\sin x + C$.
 - 9 Find $\int \sin x dx$.
 $-\cos x + C$.
 - 10 Find $\int e^x dx$.
 $e^x + C$.
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Evaluating definite integrals

- 11 Evaluate $\int_0^2 3x^2 dx$.
 $[x^3]_0^2 = 8$.
- 12 Evaluate $\int_1^3 2x dx$.
 $[x^2]_1^3 = 9 - 1 = 8$.
- 13 Evaluate $\int_0^4 x dx$.
 $\left[\frac{x^2}{2}\right]_0^4 = 8$.
- 14 Evaluate $\int_{-1}^2 3x^2 dx$.
 $[x^3]_{-1}^2 = 8 - (-1) = 9$.

Definite integrals with polynomial integrands

- 15 Evaluate $\int_0^2 (x^2 + 1) dx$.
 $\left[\frac{x^3}{3} + x\right]_0^2 = \frac{8}{3} + 2 = \frac{14}{3}$.
- 16 Evaluate $\int_1^3 (2x^2 - 3x + 1) dx$.
 $F(x) = \frac{2x^3}{3} - \frac{3x^2}{2} + x$. $F(3) = 18 - 13.5 + 3 = 7.5$. $F(1) = 0.667 - 1.5 + 1 = 0.167$. Integral
 $= 7.5 - 0.167 = \frac{22}{3}$.
- 17 Evaluate $\int_0^1 (6x^2 - 4x + 2) dx$.
 $[2x^3 - 2x^2 + 2x]_0^1 = (2 - 2 + 2) - 0 = 2$.
- 18 Evaluate $\int_{-2}^1 (x^2 - x) dx$.
 $F(x) = \frac{x^3}{3} - \frac{x^2}{2}$. $F(1) = -\frac{1}{6}$. $F(-2) = -\frac{8}{3} - 2 = -\frac{14}{3}$. Integral $= -\frac{1}{6} + \frac{14}{3} = \frac{9}{2}$.

Definite integrals of trigonometric and exponential functions

- 19 Evaluate $\int_0^{\pi/2} \cos x dx$.
 $[\sin x]_0^{\pi/2} = 1 - 0 = 1$.
- 20 Evaluate $\int_0^{\pi} \sin x dx$.
 $[-\cos x]_0^{\pi} = -\cos(\pi) - (-\cos(0)) = 1 - (-1) = 2$.
- 21 Evaluate $\int_0^1 e^x dx$.
 $[e^x]_0^1 = e - 1 \approx 1.72$.

Area under a curve

- 22 Find the area under $y = x^2$ from $x = 0$ to $x = 3$, using a definite integral.

$$\text{Area} = \int_0^3 x^2 dx = \left[\frac{x^3}{3} \right]_0^3 = \frac{27}{3} = 9.$$
- 23 Find the area under $y = 4 - x^2$ from $x = -2$ to $x = 2$.

$$F(x) = 4x - \frac{x^3}{3}. F(2) = 8 - \frac{8}{3} = \frac{16}{3}. F(-2) = -8 + \frac{8}{3} = -\frac{16}{3}. \text{Area} = \frac{16}{3} - \left(-\frac{16}{3}\right) = \frac{32}{3}.$$
- 24 Find the area under $y = 3\sqrt{x}$ from $x = 0$ to $x = 4$.

$$\int 3x^{1/2} dx = 2x^{3/2}. \text{At } x = 4: 2(8) = 16. \text{At } x = 0: 0. \text{Area} = 16.$$
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Signed area and regions below the axis

- 25 Evaluate $\int_0^2 (x - 2) dx$ and explain what the sign of the result means.

$$\left[\frac{x^2}{2} - 2x \right]_0^2 = (2 - 4) - 0 = -2. \text{The negative sign means the curve lies below the } x\text{-axis on } [0, 2].$$
- 26 Find the (positive) area between $y = x - 2$ and the x -axis, from $x = 0$ to $x = 2$.
 The signed integral is -2 (from the previous question), so the area is $|-2| = 2$.
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Properties of definite integrals

- 27 Given $\int_1^4 f(x) dx = 10$, find $\int_1^4 3f(x) dx$.
 By linearity, $\int_1^4 3f(x) dx = 3(10) = 30$.
- 28 Given $\int_0^2 f(x) dx = 5$ and $\int_2^5 f(x) dx = 7$, find $\int_0^5 f(x) dx$.
 By additivity over intervals: $5 + 7 = 12$.
- 29 Given $\int_0^3 f(x) dx = 8$, find $\int_3^0 f(x) dx$.
 Reversing the limits negates the integral: $\int_3^0 f(x) dx = -8$.
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Finding a function from its derivative

- 30 A function satisfies $f'(x) = 6x^2 - 4$ and $f(1) = 5$. Find $f(x)$.

$$f(x) = 2x^3 - 4x + C. f(1) = 2 - 4 + C = 5 \Rightarrow C = 7. f(x) = 2x^3 - 4x + 7.$$
- 31 A function satisfies $f'(x) = 3x^2 + 2x$ and $f(0) = 1$. Find $f(x)$.

$$f(x) = x^3 + x^2 + C. f(0) = C = 1. f(x) = x^3 + x^2 + 1.$$
- 32 A function satisfies $f'(x) = 4x - 1$ and $f(2) = 10$. Find $f(x)$.

$$f(x) = 2x^2 - x + C. f(2) = 8 - 2 + C = 10 \Rightarrow C = 4. f(x) = 2x^2 - x + 4.$$
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Average value of a function

- 33 Find the average value of $f(x) = x^2$ on $[0, 3]$, using $\text{avg} = \frac{1}{b-a} \int_a^b f(x) dx$.
 $\text{avg} = \frac{1}{3} \int_0^3 x^2 dx = \frac{1}{3}(9) = 3$.
- 34 Find the average value of $f(x) = 2x + 1$ on $[1, 5]$.
 $\int_1^5 (2x + 1) dx = [x^2 + x]_1^5 = 30 - 2 = 28$. $\text{avg} = \frac{28}{5-1} = 7$.
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Word problems: integration in context

- 35 This question has two parts. A particle moves along a line with velocity $v(t) = 6t - 3t^2$ (m/s) for $0 \leq t \leq 4$.
- a Find the particle's displacement from $t = 0$ to $t = 2$, using $\int_0^2 v(t) dt$.
 $\int_0^2 (6t - 3t^2) dt = [3t^2 - t^3]_0^2 = 12 - 8 = 4$ meters.
- b Find the particle's displacement from $t = 0$ to $t = 4$, and explain why it is not simply twice the part (a) answer. $\int_0^4 (6t - 3t^2) dt = [3t^2 - t^3]_0^4 = 48 - 64 = -16$ meters. The velocity becomes negative for $t > 2$ (the particle reverses direction), so the displacements over $[0, 2]$ and $[2, 4]$ partially cancel rather than simply add.
- 36 This question has two parts. Water flows into a tank at a rate of $r(t) = 20 - 2t$ liters per minute, for $0 \leq t \leq 10$.
- a Write a definite integral for the total volume of water that flows in during the first 5 minutes.
 $V = \int_0^5 (20 - 2t) dt$.
- b Evaluate the integral to find the volume. $V = [20t - t^2]_0^5 = 100 - 25 = 75$ liters.